A First Course in Abstract Algebra

Section 34: Isomorphism Theorems

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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34 Isomorphism Theorems

Exercise 34.1: A homomorphism from $\mathbb{Z}_{12}$ to $\mathbb{Z}_3$

Plan. An additive homomorphism is determined by the image of 1, so $\phi(x) = 2x \pmod 3$. **(a)** $2x \equiv 0 \pmod 3$ holds exactly when $3 \mid x$. Thus $K = \{0, 3, 6, 9\}$. **(b)** The distinct cosets are

$$K = \{0, 3, 6, 9\}, \quad 1 + K = \{1, 4, 7, 10\}, \quad 2 + K = \{2, 5, 8, 11\}.$$

(c) The first-isomorphism map is $x + K \mapsto \phi(x)$, hence $K \mapsto 0$, $1 + K \mapsto 2$, $2 + K \mapsto 1$. Changing x by an element of K does not change $\phi(x)$, so the map is well defined; it is a bijective homomorphism, not merely a matching of three sets.

Exercise 34.2: A homomorphism from $\mathbb{Z}_{18}$ to $\mathbb{Z}_{12}$

(a) Here $\phi(x) = 10x \pmod{12}$. Its kernel solves $12 \mid 10x$, equivalently $6 \mid x$, so $K = \{0, 6, 12\}$. **(b)** The six cosets are $r + K = \{r, r + 6, r + 12\}$ for $0 \le r \le 5$:

$$\{0, 6, 12\},\ \{1, 7, 13\},\ \{2, 8, 14\},\ \{3, 9, 15\},\ \{4, 10, 16\},\ \{5, 11, 17\}.$$

(c) Multiples of 10 modulo 12 give $\phi[\mathbb{Z}_{18}] = \{0, 2, 4, 6, 8, 10\}$. **(d)** Under $r + K \mapsto 10r \pmod{12}$, the cosets listed above map respectively to $0, 10, 8, 6, 4, 2$. Addition is preserved because $10(r + s) = 10r + 10s$; the kernel calculation makes the quotient map injective.

Exercise 34.3: The second isomorphism theorem in $\mathbb{Z}_{24}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(4, 6) \rangle$ and $H \cap N = \langle \operatorname{lcm}(4, 6) \rangle$. **(a)** Thus $H + N = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22\}$ and $H \cap N = \{0, 12\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 6, 12, 18\}$
$2 + N$	$\{2, 8, 14, 20\}$
$4 + N$	$\{4, 10, 16, 22\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 12\}$
$4 + I$	$\{4, 16\}$
$8 + I$	$\{8, 20\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 4 + N \longmapsto 4 + I, \quad 2 + N \longmapsto 8 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.4: The second isomorphism theorem in $\mathbb{Z}_{36}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(6, 9) \rangle$ and $H \cap N = \langle \operatorname{lcm}(6, 9) \rangle$. **(a)** Thus $H + N = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33\}$ and $H \cap N = \{0, 18\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 9, 18, 27\}$
$3 + N$	$\{3, 12, 21, 30\}$
$6 + N$	$\{6, 15, 24, 33\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 18\}$
$6 + I$	$\{6, 24\}$
$12 + I$	$\{12, 30\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 6 + N \longmapsto 6 + I, \quad 3 + N \longmapsto 12 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.5: The third isomorphism theorem in $\mathbb{Z}_{24}$

(a) $H = \langle 4 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 4 \rangle$	$\{0, 4, 8, 12, 16, 20\}$
$1 + \langle 4 \rangle$	$\{1, 5, 9, 13, 17, 21\}$
$2 + \langle 4 \rangle$	$\{2, 6, 10, 14, 18, 22\}$
$3 + \langle 4 \rangle$	$\{3, 7, 11, 15, 19, 23\}$

(b) $K = \langle 8 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 8 \rangle$	$\{0, 8, 16\}$
$1 + \langle 8 \rangle$	$\{1, 9, 17\}$
$2 + \langle 8 \rangle$	$\{2, 10, 18\}$
$3 + \langle 8 \rangle$	$\{3, 11, 19\}$
$4 + \langle 8 \rangle$	$\{4, 12, 20\}$
$5 + \langle 8 \rangle$	$\{5, 13, 21\}$
$6 + \langle 8 \rangle$	$\{6, 14, 22\}$
$7 + \langle 8 \rangle$	$\{7, 15, 23\}$

(c) In H/K there are two elements, namely K and $4 + K$; their underlying sets are

$$K = \{0, 8, 16\}, \qquad 4 + K = \{4, 12, 20\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+4}\} \qquad (0 \le r < 4).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets.
(e) The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+4}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.6: The third isomorphism theorem in $\mathbb{Z}_{36}$

(a) $H = \langle 9 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 9 \rangle$	$\{0, 9, 18, 27\}$
$1 + \langle 9 \rangle$	$\{1, 10, 19, 28\}$
$2 + \langle 9 \rangle$	$\{2, 11, 20, 29\}$
$3 + \langle 9 \rangle$	$\{3, 12, 21, 30\}$
$4 + \langle 9 \rangle$	$\{4, 13, 22, 31\}$
$5 + \langle 9 \rangle$	$\{5, 14, 23, 32\}$
$6 + \langle 9 \rangle$	$\{6, 15, 24, 33\}$
$7 + \langle 9 \rangle$	$\{7, 16, 25, 34\}$
$8 + \langle 9 \rangle$	$\{8, 17, 26, 35\}$

(b) $K = \langle 18 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 18 \rangle$	$\{0, 18\}$
$1 + \langle 18 \rangle$	$\{1, 19\}$
$2 + \langle 18 \rangle$	$\{2, 20\}$
$3 + \langle 18 \rangle$	$\{3, 21\}$
$4 + \langle 18 \rangle$	$\{4, 22\}$
$5 + \langle 18 \rangle$	$\{5, 23\}$
$6 + \langle 18 \rangle$	$\{6, 24\}$
$7 + \langle 18 \rangle$	$\{7, 25\}$
$8 + \langle 18 \rangle$	$\{8, 26\}$
$9 + \langle 18 \rangle$	$\{9, 27\}$
$10 + \langle 18 \rangle$	$\{10, 28\}$
$11 + \langle 18 \rangle$	$\{11, 29\}$
$12 + \langle 18 \rangle$	$\{12, 30\}$
$13 + \langle 18 \rangle$	$\{13, 31\}$
$14 + \langle 18 \rangle$	$\{14, 32\}$
$15 + \langle 18 \rangle$	$\{15, 33\}$
$16 + \langle 18 \rangle$	$\{16, 34\}$
$17 + \langle 18 \rangle$	$\{17, 35\}$

(c) In H/K there are two elements, namely K and $9 + K$; their underlying sets are

$$K = \{0, 18\}, \qquad 9 + K = \{9, 27\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+9}\} \qquad (0 \le r < 9).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets. **(e)** The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+9}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.7: Normality of an intersection

Let $h \in H$ and $x \in H \cap N$. Since H is a subgroup, $hxh^{-1} \in H$. Since $N \trianglelefteq G$ and $h \in G$, also $hxh^{-1} \in N$. Thus $h(H \cap N)h^{-1} \subseteq H \cap N$. Apply the same argument to h^{-1} to obtain the reverse inclusion. Hence $H \cap N \trianglelefteq H$, directly by the conjugation definition of normality.

Exercise 34.8: Successive quotient groups

(a) For $gH \in A$ and $kH \in B$, $(gH)(kH)(gH)^{-1} = (gkg^{-1})H \in B$ because $K \trianglelefteq G$. This proves $B \trianglelefteq A$; replacing K by L proves $C \trianglelefteq A$. Also $K \subseteq L$ implies $B \subseteq C$. If the original inclusion is strict, so is this one: $l \in L \setminus K$ gives $lH \in C \setminus B$. **(b)** The third isomorphism theorem gives $A/B \cong G/K$ and identifies C/B with L/K. Therefore

$$(A/B)/(C/B) \cong (G/K)/(L/K) \cong G/L.$$

Explicitly the final map sends the class represented successively by gH, then by $(gH)B$, to gL. Its kernel consists precisely of representatives in L.

Exercise 34.9: Two complementary normal subgroups

Because $K, L \trianglelefteq G$, the product KL is a subgroup containing both; hence $KL = K \vee L = G$. Restrict the quotient map $G \to G/K$ to L. Every $g = kl$ has $gK = lK$ (normality permits changing kl to lk'), so $l \mapsto lK$ is onto. Its kernel is $L \cap K = \{e\}$, so it is an isomorphism $L \cong G/K$. Interchanging K and L gives $K \cong G/L$. These conclusions require normality, not merely trivial intersection.